THE FP FLAT PROPER SPACES STACK SATISFIES (RS*) ON PUSHOUTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma more-morphisms-lemma-pushout-along-thickening (canonical tag 07RT). cited at proof line 3773. The pushout square.
- Lemma spaces-pushouts-lemma-equivalence-categories-spaces-pushout-flat (canonical tag 07W3). cited at proof line 3791. The equivalence without properness.
- Lemma spaces-morphisms-lemma-base-change-proper (canonical tag 04WP). cited at proof line 3799. Properness preserved under base change.
- Lemma spaces-more-morphisms-lemma-thicken-property-morphisms-cartesian (canonical tag 09ZZ). cited at proof line 3803. Properness detected on the thickening pullback.

Source reference. Original lemma 07RT: sources/stacks-chow/tags.txt, source lines 9878–9878. Cited at proof line 3773: the pushout square.

Source reference. Original lemma 07W3: sources/stacks-chow/tags.txt, source lines 10028–10028. Cited at proof line 3791: the equivalence without properness.

Source reference. Original lemma 04WP: sources/stacks-chow/tags.txt, source lines 6367–6367. Cited at proof line 3799: properness preserved.

Source reference. Original lemma 09ZZ: sources/stacks-chow/tags.txt, source lines 12617–12617. Cited at proof line 3803: properness detected on the thickening pullback.

Lemma 0.2. *Let*

$$\begin{array}{ccc} T & \longrightarrow & T' \\ \downarrow & & \downarrow \\ S & \longrightarrow & S' \end{array}$$

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be a pushout in the category of schemes where $T \to T'$ is a thickening and $T \to S$ is affine, see More on Morphisms, Lemma 07RT. Then the functor on fibre categories

$$\begin{array}{c} \mathcal{S}paces'_{fp,flat,proper,S'} \\ \downarrow \\ \mathcal{S}paces'_{fp,flat,proper,S} \times_{\mathcal{S}paces'_{fp,flat,proper,T}} \mathcal{S}paces'_{fp,flat,proper,T'} \end{array}$$

is an equivalence.

Proof. The functor is an equivalence if we drop "proper" from the list of conditions and replace "of finite presentation" by "locally of finite presentation", see Pushouts of Spaces, Lemma 07W3. Thus it suffices to show that given a morphism $X' \to S'$ of an algebraic space to S' which is flat and locally of finite presentation, then $X' \to S'$ is proper if and only if $S \times_{S'} X' \to S$ and $T' \times_{S'} X' \to T'$ are proper. One implication follows from the fact that properness is preserved under base change (Morphisms of Spaces, Lemma 04WP) and the other from the fact that properness of $S \times_{S'} X' \to S$ implies properness of $X' \to S'$ by More on Morphisms of Spaces, Lemma 09ZZ. □